

Round 157 Pentad Discovery (Backbone-First): (1) $\Delta x(\text{SGR1745}) = \text{SO}_5^{-10}$ m Atomic-Scale Length 3RD-Instance of SO_5^{-10} 3-Domain Family (Length + Magnetic-Field + Amplitude) + (2) $I(\text{Tapestry}) = \text{SO}_5^{20}$ A Current-Domain 2ND-Instance with Sgr A* SO_5^{24} + (3) $f_{\text{DPM}} = \text{SO}_5^{11}$ Hz Frequency Slot $n=11$ + (4) $f_{\text{DPM}} = \text{SO}_5^{12}$ Hz Frequency Slot $n=12$ + (5) $f_{\text{aether}} = \text{SO}_5^4$ Hz Aether-Frequency Slot $n=4$; Plus CROSS-OBJECT $I(\text{aether-resonance}) = \text{SO}_5^{21}$ A 4TH-INSTANCE of PAPER_1955/1991/1993 3-Class Family

Daniel T. Murphy

PAPER_2023 — Round 157 Pentad Discovery (Backbone-First Discipline)

Author: Daniel T. Murphy | **Framework:** UQFF Star-Magic v5.66+ | **Date:** 2026-07-14

Motivation — 16th Consecutive Backbone-First Round

R157 initial fills identified 6 tentative novelties. Backbone-first double-check confirms 5 novel and correctly withdraws 1 ($I = \text{SO}_5^{21}$ A) as 4th-instance of already-documented 3-class family.

Abstract

Discovery 1 — $\Delta x(\text{SGR1745}) = \text{SO}_5^{-10}$ m Atomic-Scale Length 3RD-Instance:

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$\Delta x(\text{SGR1745 quantum uncertainty length scale}) = 1\text{e-}10 \text{ m} = \text{SO}_5^{-10} \text{ m EXACT}$

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Novel 3RD-instance of SO_5^{-10} multi-dimensional family:

- Length (this): SGR1745 quantum-uncertainty Δx
- Magnetic-field: Saturn B (PAPER_2019 R153 D4)
- Amplitude: M16 A_{osc} (PAPER_2022 R156 D1 — algebraically = F_{TRZ}^{10} per PAPER_1960 = SO_5^{-10})

Three-domain application of SO_5^{-10} primitive-composed exponent.

Discovery 2 — $I(\text{Tapestry NGC 2014/2020}) = \text{SO}_5^{20}$ A Current-Domain 2ND-Instance:

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$I(\text{Tapestry cluster wind current}) = 1\text{e}20 \text{ A} = \text{SO}_5^{20} \text{ A EXACT}$

``

Novel current-domain 2ND-instance:

- SO_5^{21} A: Sgr A* SMBH DPM current (PAPER_1994 R132 seminal)

- **SO_5² A: Tapestry cluster wind current** (this)

Now 2-object current-domain family at n=20 and n=24.

Discovery 3 — f_DPM = SO_5¹¹ Hz Frequency Slot n=11:

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f_DPM(Tapestry NGC 2014/2020) = 1e11 Hz = SO_5¹¹ Hz EXACT

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Novel frequency-domain SO_5-power slot at n=11 (parallel to SO_5¹¹ mass 3-prefix family PAPER_2011 R148 D1).

Discovery 4 — f_DPM = SO_5¹² Hz Frequency Slot n=12:

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f_DPM(General aether resonance) = 1e12 Hz = SO_5¹² Hz EXACT

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Novel frequency-domain SO_5-power slot at n=12 (parallel to SO_5¹² mass PAPER_2012 R149 fill 1 CenA primary).

Discovery 5 — f_aether = SO_5⁴ Hz Aether-Frequency Slot n=4:

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f_aether(General aether resonance) = 1e4 Hz = SO_5⁴ Hz EXACT

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Novel aether-frequency SO_5-power slot at n=4. First documented aether-frequency slot in SO_5-power ladder.

Cross-Object Confirmation (Withdrawn from Novelty)

I(aether-resonance) = SO_5²¹ A — CROSS-OBJECT 4TH-INSTANCE of already-established 3-class SO_5²¹ A current-domain family. Prior seminal documentation:

- **PAPER_1955** — "10²¹ A and both lock onto SO_5²¹ A in the same round from independent CP1 classes"

- **PAPER_1991** — "SGR1745FreqDPMCalculator (I = 10²¹ A) and [C-class] SO_5²¹ A twin-instance at same round"

- **PAPER_1993** — "SO_5²¹ A three-class family extension"

R157 Fill 5 ResonanceAetherCalculator I = 10²¹ A is the 4TH-INSTANCE of this documented family, not novel. Correctly attributed to prior seminal papers.

All 5 discoveries + 1 confirmation validated at fidelity gate 1286/0.

SO_5 Current-Domain Ladder (Post PAPER_2023)

| n | Value (A) | Object | Attribution |

---|---|---|---

| 20 | 1e20 | Tapestry NGC 2014/2020 | PAPER_2023 R157 D2 NEW |

| 21 | 1e21 | SGR1745FreqDPM + siblings (3-class + ResonanceAether 4th) | PAPER_1955/1991/1993 seminal + R157 confirm |

| 24 | 1e24 | Sgr A* SMBH DPM | PAPER_1994 R132 seminal |

Now 3-slot ladder in current dimensional domain.

SO_5 Frequency-Domain Ladder Extensions

n	Value (Hz)	Object	Attribution
4	1e4	General aether-frequency	PAPER_2023 R157 D5 NEW
11	1e11	Tapestry f_DPM	PAPER_2023 R157 D3 NEW
12	1e12	General aether f_DPM	PAPER_2023 R157 D4 NEW

Frequency ladder extended by 3 new slots.

SO_5^-10 3-Domain Family (Post PAPER_2023)

Domain	Object	Attribution
Length (m)	SGR1745 Δx	PAPER_2023 R157 D1 NEW
Magnetic-field (T)	Saturn B	PAPER_2019 R153 D4
Amplitude (dimensionless)	M16 A_osc = F_TRZ^10	PAPER_2022 R156 D1 (algebraic via PAPER_1960)

3-orthogonal-domain family at n=-10 exponent established.

Wiring Plan (5 dispatches, lowercase keys)

- delta_x_sgr1745_so_5_neg_10_atomic_length → 1e-10
- i_tapestry_so_5_20_current_domain → 1e20
- f_dpm_so_5_11_frequency_slot → 1e11
- f_dpm_so_5_12_frequency_slot → 1e12
- f_aether_so_5_4_frequency_slot → 1e4

Cross-References

- PAPER_1955 — I = SO_5^21 A seminal 3-class family (Cross-obj attribution source)
- PAPER_1990 — SO_5-power frequency ladder (D3/D4/D5 basis)
- PAPER_1991 — SGR1745FreqDPM SO_5^21 A twin (Cross-obj source)
- PAPER_1993 — SO_5^21 A three-class family (Cross-obj source)
- PAPER_1994 — R132 Sgr A* SMBH SO_5^24 A current-domain seminal (D2 basis)
- PAPER_1960 — F_TRZ = 1/SO_5 (D1 algebraic equivalence basis for M16 A entry)
- PAPER_2019 — R153 D4 Saturn B SO_5^-10 (D1 sibling)
- PAPER_2022 — R156 D1 M16 A SO_5^-10 (D1 sibling)

Conclusion

R157 16th-consecutive backbone-first round yields **5 novel discoveries + 1 correctly-attributed cross-object confirmation**.

Cumulative through PAPER_2023: 90 first-pass novel + 21 confirmations + 3 audit sweeps + 4 self-corrections + 1 backbone-recovery + 2 equivalence-restatement withdrawals + 1 family-extension attribution.

End of PAPER_2023 Draft 1.